MICROBURBS HACKATHON · TASK 4 · TEAM-OPAL

Would you let one flat vote ten times?

We price every City of Sydney flat from its own past rent and its neighbours in the same building — one flat, one vote — and only trust evidence that is still fresh.

$45
typical miss on 2,898 held-out adverts (Jun–Jul 2026)

77%
within $100 a week

0.88
R^2 on the same adverts

65,777
flats priced, with and without parking

Organisers' reference points on the scored Aug–Sep adverts: simple estimate $100, with parking $93, Microburbs rental valuation $86 (55% within $100). Different adverts — the official score comes from Aug–Sep.

THE PROBLEM

Same building, same bedrooms — up to $500 apart

45 Shelley St, Sydney · one-bedroom flats, latest advert since 2025

floor	water in sight	asking rent
2–3	0% of directions	$700 – $875
4–5	3–17%	$685 – $850
6–7	**25–33%**	**$875 – $1,200**

View per floor measured with NSW LiDAR line-of-sight (36 directions, 1.5 km).

Suburb averages can't see this.

Government bond data by postcode × bedrooms misses by $100 a week.

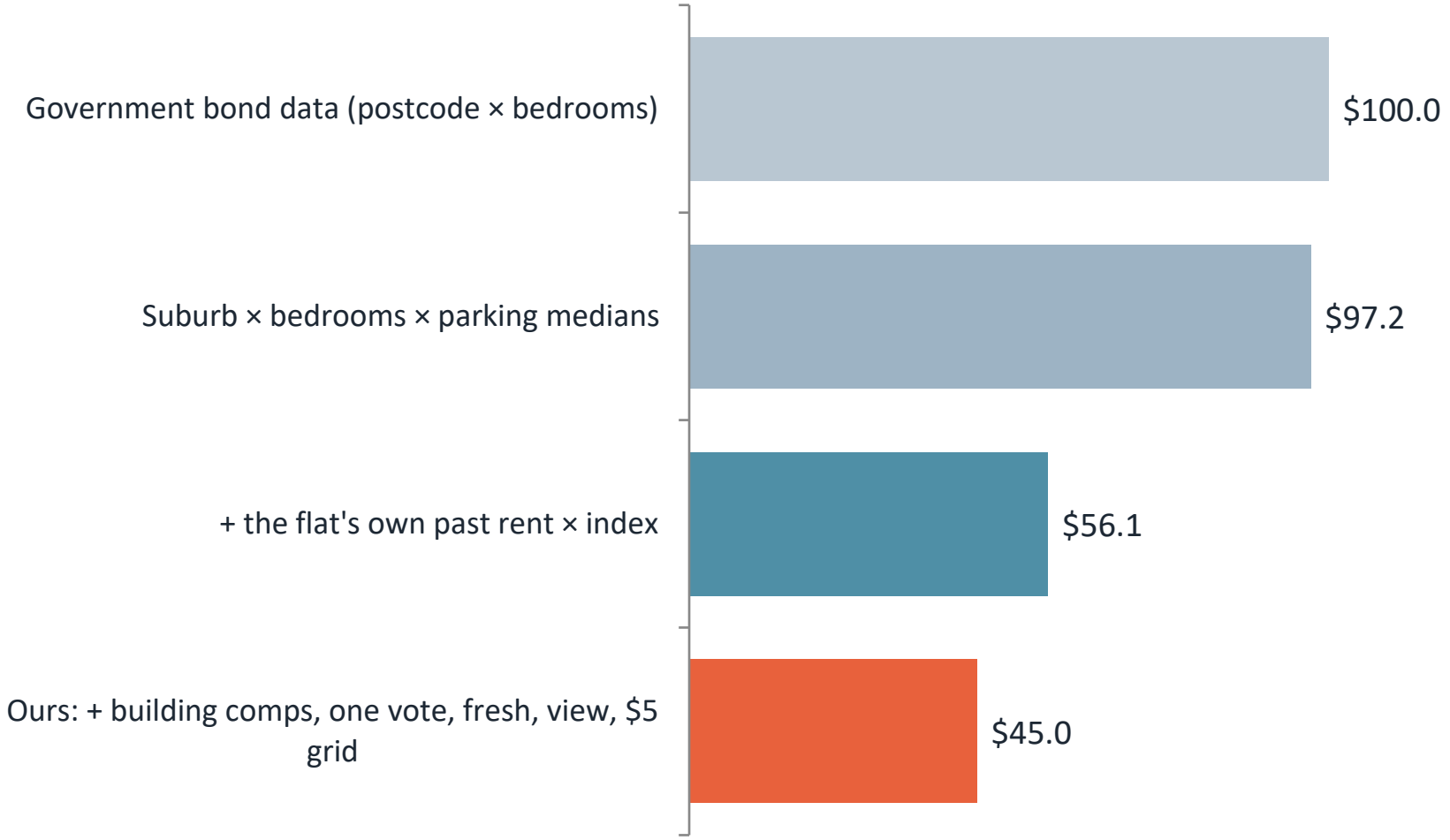
The flat itself can.

Its past rent and its neighbours' rents already price the floor, the view, the noise and the renovation.

The hard part is which evidence to trust:

stale adverts, relisted flats, changed bedroom counts.

From $100 to $45 — each step on the same adverts



Share within $100

51% → 53% → 69% → 77%

Window A (May) shows the same ladder: $100 → $92 → $57 → $45.

Bond files restricted to those published before each cut-off.

METHOD

Four steps, rebuilt from adverts before each cut-off

1 Repeat-rent index

Pairs of adverts of the same flat give the market move — location and quality cancel out. Held flat after July.

2 Own history

The flat's latest advert with the same bedroom count, moved to today. Older than 3 years → dropped as stale.

3 Building comps, one flat one vote

Same building and bedrooms; each neighbour counts once however often it relisted; shrunk to suburb medians when few.

4 View and the grid

LiDAR floor-view difference against those neighbours; parking premium for the second column; snapped to $5.

Every feature rebuilt per window — and what it did

feature	built from	how	effect
advert table + building key	past leases + API; building_address + suburb	153,040 adverts; 3,671 buildings	used
repeat-rent index	pairs of adverts of the same flat	60-day lettings; monthly log index, smoothed	used
parking premium	with vs without, suburb × bedrooms	median 8%, clipped 0–35%	used
suburb ladder	last 12 months, index-adjusted	suburb × bed × parking → … → city	used
own history + stale flag	latest same-bedroom advert	> 36 months old → not trusted	$53 → $46.5 (A)
one-vote building comps	same building + bedrooms, 36 months	one signal per neighbour, target out, n/(n+0.5)	no-history $99 → $60 (B)
floor-view difference	LiDAR view vs comps' mean view	within-building coefficients	R^2 0.877 → 0.880 (B)
price grid	predictions	nearest $5, then $0.50	$46.5 → $45 (A)

Dropped on window A: plain comps, other shrinkage, 24/60-month stale rules, $10 rounding, building-level public layers ($45 → $50), floor number instead of LiDAR.
Kept out (leakage): first_ever_listed, attrs_last_seen, API leased flag, rent-map.

Two laser surveys → a view test for every floor

2

surveys: Apr 2013, May 2020

5.6 GB

80 files of point clouds + 1 m terrain

4,399

buildings measured (99.9%)

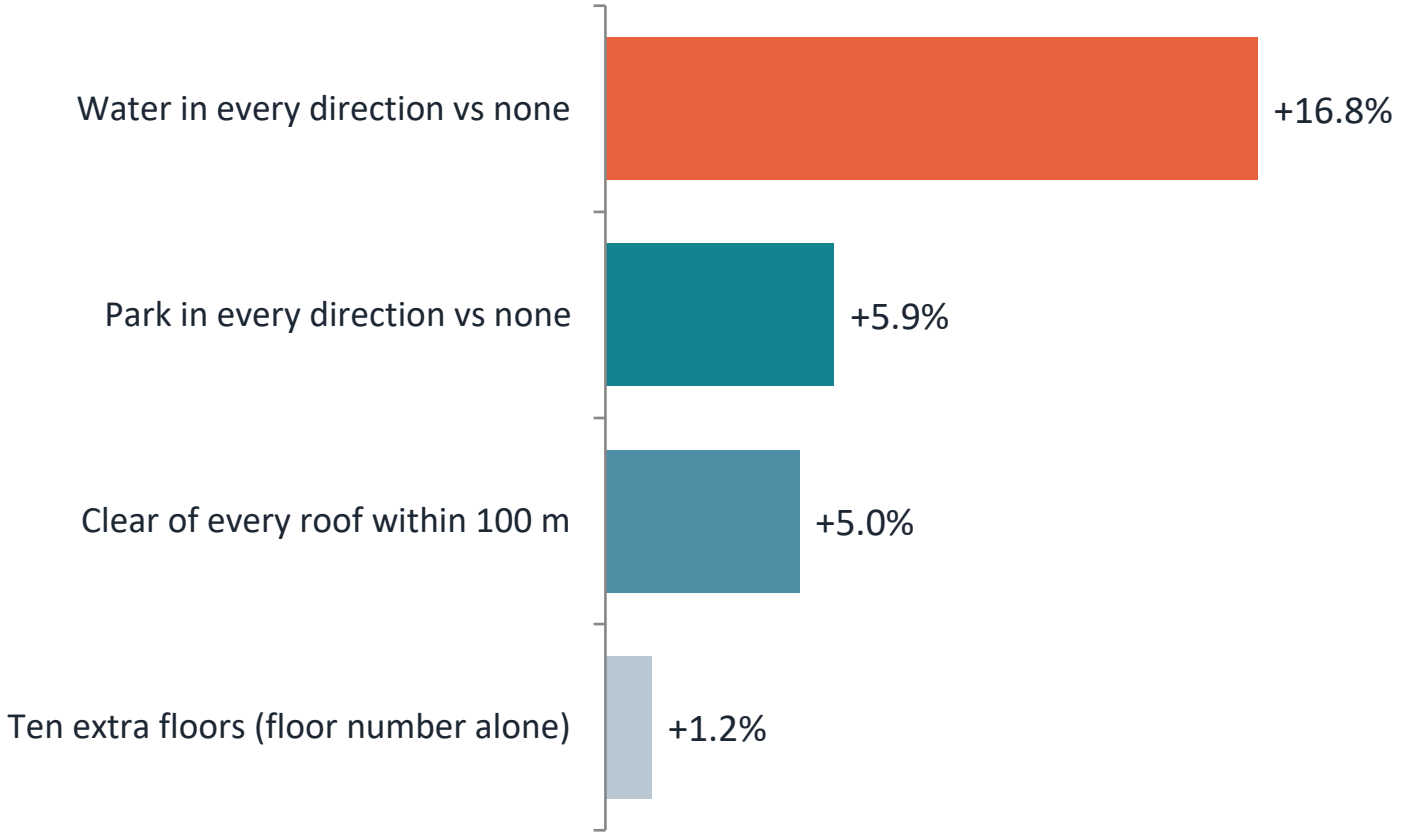
29,970

building × floor rows

1. **Height grid** DSM (highest return) − DEM (bare earth) = height above ground, 1 m cells
2. **Building height** 90th percentile inside each footprint; levels within 1 floor of OSM
3. **New builds** < 3 m in 2013, ≥ 9 m in 2020 → 79 buildings
4. **View per floor** eye at 3.1 m × floor + 1.5 m; 36 directions, 5 m steps, 1.5 km
5. **Three features** share of directions seeing water · park · above every roof within 100 m
6. **Join to flats** building address → footprint; floor_level (52%) or building median

Limits: unit orientation unknown (a floor's view is an upper bound); buildings finished after May 2020 have no 2020 height.

It's the view, not the floor



Fixed-effects estimate, window B: 22,905 flats in 2,912 building × bedroom groups (window A +16.3%, final +18.7%).

Floor number vs LiDAR

Same-building adjustment, R^2:

none 0.877 · floor 0.878 · LiDAR 0.880

LiDAR covers 22,593 flats; floor number 15,516 (48% have none).

A high floor behind a tower sees nothing; a low floor on the water sees the harbour. Only the line of sight tells them apart.

Building-level layers made errors worse

Tested and rejected on window A

- Against suburb medians: water view +25%, park +11%, strata age −6% per decade, flight path −2%, new build −4%
- Added on top of building comps: overall $45 → $50, no-history $65 → $74.5
- Why: the neighbours' rents already contain the building's view, age and noise — counting them again double-counts

Kept

- Only the floor-view difference between a flat and the neighbours it is compared with — the one thing their rents cannot contain
- Heat, canopy, flight path: insight maps; after the model their residuals sit within ±2%
- Window A share within $100 79.1% → 79.2%; window B neutral at $45, R^2 0.877 → 0.880

Three non-obvious findings

Old ≠ useful

A flat's own advert from more than 3 years ago is worse than its neighbours' current rents.

$53 → $46.5

on window A when swapped

Same address ≠ same flat

When the bedroom count changes between adverts (whole-building ids, split lettings) the error multiplies.

× 2.4

error when bedrooms differ

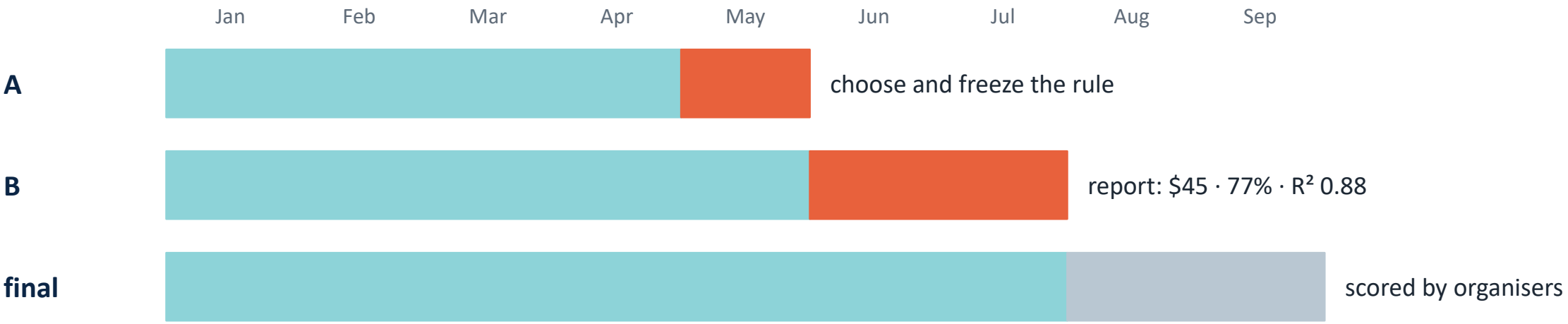
Rents live on a grid

84% of asking rents end in $0, 52% are multiples of $50, 26% of $100.

$46.5 → $45

on window A after snapping to $5

Built only on earlier adverts, tested on later ones



- Train bars = index, medians, comps and parking premium rebuilt from that window's adverts only; the code refuses any training row dated after the cut-off.
- Candidates (one-vote strength, stale threshold, snapping, public layers) compared on A only; B evaluated once with the frozen rule. The floor-view step was added on A after the base rule froze — both B results are reported.
- Our estimates sit about 1% above later adverts — consistent with Sydney unit asking rents flattening after July.

A rent you can check, flat by flat

Tenants

see whether an asking rent is high for this floor of this building

Landlords

set a rent that fits the flats around them, not the suburb average

Investors

compare likely rent before they buy

Next: unit orientation, strata sale prices within the building, and an evidence card per estimate — own advert, neighbours used, view adjustment.

github.com/HollyHsu0808/team-opal-rents

rents_team-opal.csv · report.html · FEATURES.md · run_everything.py

Data: Microburbs hackathon files and API (used on the day, not republished) · NSW Spatial Services LiDAR, cadastre, boundaries (CC BY) · OpenStreetMap (ODbL) · NSW DPHI heat and canopy (CC BY 4.0) · NSW DCCEEW urban heat island (CC BY) · ABS Census 2021 (CC BY 4.0) · NSW Fair Trading rental bonds (baseline only)